

# คณะวิศวกรรมศาสตร์ มหาวิทยาลัยหอการค้าไทย

## ผศ.เฉลิมชนม์ ไวศยดำรง

- **NL217** โปรแกรมคอมพิวเตอร์ในการแก้ปัญหาทางวิศวกรรม  
<https://github.com/chalermchonvis/>

|                                                                                                                               |                   |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Description CLASSICI<br>TORTELLONIALLA RICOTTA E SPINACI                                                                      |                   |
| Article Code<br>RAG040                                                                                                        | Weight<br>250g    |
| GTIN number<br>09012348410268                                                                                                 | N° Pck<br>2       |
| Production date<br>16/02/10                                                                                                   | Lott no<br>012345 |
| <br>(01)09012348410268(15)100816(10)012345 |                   |

โหลดเอกสารและไฟล์ที่ใช้ในการสอนที่ Subject -> NL344

[Link](#)

<https://github.com/chalermchonvis/UTCC-Engineering-Subjects/tree/main>

[Downlodin Libre Barcode 39 Font](#)

<https://fonts.google.com/specimen/Libre+Barcode+39?preview.script=Latn>

## Relative cell references

| B2 |          | ✖ ✓ <i>fx</i> | =C2*B2    |        |   |   |
|----|----------|---------------|-----------|--------|---|---|
|    | A        | B             | C         | D      | E | F |
| 1  | Trainers | Price         | Pokeballs | Cost   |   |   |
| 2  | Iva      | 2             | 2         | =C2*B2 |   |   |
| 3  | Liam     | 2             | 5         |        |   |   |
| 4  | Pablo    | 2             | 10        |        |   |   |
| 5  | Jenny    | 2             | 7         |        |   |   |
| 6  | Iben     | 2             | 6         |        |   |   |
| 7  | Kasper   | 2             | 3         |        |   |   |
| 8  |          |               |           |        |   |   |

## Absolute cell references ( \$ , \$ )

| C2 |          | <i>fx</i> | =\$B\$11 |        |  |  |
|----|----------|-----------|----------|--------|--|--|
|    | A        | B         | C        |        |  |  |
| 1  | Trainers | Pokeballs | Cost     |        |  |  |
| 2  | Iva      |           | 2        | =B\$11 |  |  |
| 3  | Liam     |           | 5        |        |  |  |
| 4  | Pablo    |           | 10       |        |  |  |
| 5  | Jenny    |           | 7        |        |  |  |
| 6  | Iben     |           | 6        |        |  |  |
| 7  | Kasper   |           | 3        |        |  |  |
| 8  |          |           |          |        |  |  |
| 9  |          |           |          |        |  |  |
| 10 |          |           |          |        |  |  |
| 11 | Price    | 2         |          |        |  |  |

## Lab1 (1.1-1.4)

|   | A        | B     | C         | D    | E        |
|---|----------|-------|-----------|------|----------|
| 1 | Trainers | Price | Pokeballs | Cost | TotalSum |
| 2 | Iva      | 2     | 2         | 4    | 4        |
| 3 | Liam     | 2     | 5         | 10   | 14       |
| 4 | Pablo    | 2     | 10        | 20   | 34       |
| 5 | Jenny    | 2     | 7         | 14   | 48       |
| 6 | Iben     | 2     | 6         | 12   | 60       |
| 7 | Kasper   | 2     | 3         | 6    | 66       |

# Lab1\_1

B2 ✖ ✓ *fx* =C2\*B2

|   | A        | B     | C         | D      | E | F |
|---|----------|-------|-----------|--------|---|---|
| 1 | Trainers | Price | Pokeballs | Cost   |   |   |
| 2 | Iva      | 2     | 2         | =C2*B2 |   |   |
| 3 | Liam     | 2     | 5         |        |   |   |
| 4 | Pablo    | 2     | 10        |        |   |   |
| 5 | Jenny    | 2     | 7         |        |   |   |
| 6 | Iben     | 2     | 6         |        |   |   |
| 7 | Kasper   | 2     | 3         |        |   |   |
| 8 |          |       |           |        |   |   |

|   | A        | B     | C         | D    | E        |
|---|----------|-------|-----------|------|----------|
| 1 | Trainers | Price | Pokeballs | Cost | TotalSum |
| 2 | Iva      | 2     | 2         | 4    | 4        |
| 3 | Liam     | 2     | 5         | 10   | 14       |
| 4 | Pablo    | 2     | 10        | 20   | 34       |
| 5 | Jenny    | 2     | 7         | 14   | 48       |
| 6 | Iben     | 2     | 6         | 12   | 60       |
| 7 | Kasper   | 2     | 3         | 6    | 66       |

D3 ✖ ✓ *fx* =E2+D3

|   | A        | B     | C         | D    | E        | F |
|---|----------|-------|-----------|------|----------|---|
| 1 | Trainers | Price | Pokeballs | Cost | TotalSum |   |
| 2 | Iva      | 2     | 2         | 4    | 4        |   |
| 3 | Liam     | 2     | 5         | 10   | =E2+D3   |   |
| 4 | Pablo    | 2     | 10        | 20   |          |   |
| 5 | Jenny    | 2     | 7         | 14   |          |   |
| 6 | Iben     | 2     | 6         | 12   |          |   |
| 7 | Kasper   | 2     | 3         | 6    |          |   |

| B2 |          |           |              | = \$B\$11*B2 |  |  |  |
|----|----------|-----------|--------------|--------------|--|--|--|
|    | A        | B         | C            | D            |  |  |  |
| 1  | Trainers | Pokeballs | Cost         |              |  |  |  |
| 2  | Iva      | 2         | = \$B\$11*B2 |              |  |  |  |
| 3  | Liam     | 5         |              |              |  |  |  |
| 4  | Pablo    | 10        |              |              |  |  |  |
| 5  | Jenny    | 7         |              |              |  |  |  |
| 6  | Iben     | 6         |              |              |  |  |  |
| 7  | Kasper   | 3         |              |              |  |  |  |
| 8  |          |           |              |              |  |  |  |
| 9  |          |           |              |              |  |  |  |
| 10 |          |           |              |              |  |  |  |
| 11 | Price    | 2         |              |              |  |  |  |

## Lab1\_2

|    | A        | B         | C    |
|----|----------|-----------|------|
| 1  | Trainers | Pokeballs | Cost |
| 2  | Iva      | 2         | 4    |
| 3  | Liam     | 5         | 10   |
| 4  | Pablo    | 10        | 20   |
| 5  | Jenny    | 7         | 14   |
| 6  | Iben     | 6         | 12   |
| 7  | Kasper   | 3         | 6    |
| 8  |          |           |      |
| 9  |          |           |      |
| 10 |          |           |      |
| 11 | Price    | 2         |      |

Fix Price

Now() ,Today()

=IF(A2<>"",NOW(),"")

The screenshot shows an Excel spreadsheet with the following data:

|    | A             | B                  |
|----|---------------|--------------------|
| 1  | DATA          | DateTimeIN         |
| 2  | 8851234502132 | 5/10/2025 12:41 น. |
| 3  |               |                    |
| 4  |               |                    |
| 5  |               |                    |
| 6  |               |                    |
| 7  |               |                    |
| 8  |               |                    |
| 9  |               |                    |
| 10 |               |                    |
| 11 |               |                    |
| 12 |               |                    |

The formula bar shows the formula: `=IF(A2<>"",NOW(),"")`

The Format Cells dialog box is open, showing the Date category selected. The Type list includes various date and time formats, with "14/3/2012 13:30 น." selected. The Locale (location) is set to Thai.



Paste

Clipboard

Calibri

11

A<sup>^</sup>**B***I*U**B***I*U

Font



Alignment



Conditional

## Format Cells

Number

Alignment

Font

Border

Fill

Protection

## Category:

General  
Number  
Currency  
Accounting  
Date  
Time  
Percentage  
Fraction  
Scientific  
Text  
Special  
Custom

Sample

01/01/2025 5:05

Type:

dd/mm/yyyy h:mm

dd-mmm

mmm-yy

h:mm AM/PM

h:mm:ss AM/PM

h:mm

h:mm:ss

dd/mm/yyyy h:mm

mm:ss

mm:ss.0

@

[h]:mm:ss

-B\* #,##0\_-;-B\* #,##0\_-;-B\* \*-;-;-@\_-

Delete

123

General

No specific format

12

Number



Currency



Accounting



Short Date



Long Date



Time



Percentage



Fraction

More Number Formats...



# Lab 1.3

|    | A                | B                | C           | D                | E       |
|----|------------------|------------------|-------------|------------------|---------|
| 1  | Time             | Time             | Time        | Hour             |         |
| 2  | 5:30             | 20:05            | 14:35       | 15.00            |         |
| 3  |                  |                  | =B2-A2      | =ROUND(C2*24, 0) |         |
| 4  | DateTime         | DateTime         | Hour        |                  |         |
| 5  | 01/01/2025 5:05  | 02/01/2025 1:08  | 20.05       |                  |         |
| 6  |                  |                  | =(B5-A5)*24 |                  |         |
| 7  |                  |                  |             |                  |         |
| 8  | คำนวณค่า OT      | 45               | บาท/hr      |                  |         |
| 9  |                  |                  |             |                  |         |
| 10 | DateTime         | DateTime         | Hours       | OT_Cost          |         |
| 11 | 01/01/2025 5:05  | 01/01/2025 15:08 | 10.05       | 450              | 10 * 45 |
| 12 | 01/01/2025 5:05  | 01/01/2025 10:08 | 5.05        | 225              | 5*45    |
| 13 | 01/01/2025 5:05  | 01/01/2025 11:08 | 6.05        | 270              |         |
| 14 | 01/01/2025 15:05 | 02/01/2025 1:08  | 10.05       | 450              |         |
| 15 | 01/01/2025 16:05 | 02/01/2025 2:08  | 10.05       | 450              |         |
| 16 | 01/01/2025 20:05 | 02/01/2025 1:08  | 5.05        | 225              |         |
| 17 | 01/01/2025 21:05 | 02/01/2025 2:08  | 5.05        | 225              |         |



## SUMIF, AVERAGEIF, and COUNTIF

|   | A | B | C | D                       |
|---|---|---|---|-------------------------|
| 1 |   |   |   |                         |
| 2 | 1 |   | 8 | =SUMIF(A2:A9,"> 3")     |
| 3 | 2 |   |   |                         |
| 4 | 3 |   | 2 | =COUNTIF(A2:A9,"> 3")   |
| 5 | 4 |   |   |                         |
| 6 | 1 |   | 4 | =AVERAGEIF(A2:A9,"> 3") |
| 7 | 2 |   |   |                         |
| 8 | 3 |   |   |                         |
| 9 | 4 |   |   |                         |

หาค่า **OT** รวม ที่เกิน 8 ชั่วโมง  
(ชั่วโมงละ 40 บาท)

|   | A              | B              | C     |
|---|----------------|----------------|-------|
| 1 | DateTime       | DateTime       | Hours |
| 2 | 1/1/2025 5:05  | 1/1/2025 15:08 | 10.05 |
| 3 | 1/1/2025 5:05  | 1/1/2025 10:08 | 5.05  |
| 4 | 1/1/2025 5:05  | 1/1/2025 11:08 | 6.05  |
| 5 | 1/1/2025 15:05 | 2/1/2025 1:08  | 10.05 |
| 6 | 1/1/2025 16:05 | 2/1/2025 2:08  | 10.05 |
| 7 | 1/1/2025 20:05 | 2/1/2025 1:08  | 5.05  |
| 8 | 1/1/2025 21:05 | 2/1/2025 2:08  | 5.05  |

## การวิเคราะห์แบบเอบีซี ABC Analysis

[https://sc2.kku.ac.th/stat/statweb/images/Eventpic/60/Seminar/02\\_15\\_.pdf](https://sc2.kku.ac.th/stat/statweb/images/Eventpic/60/Seminar/02_15_.pdf)

<https://www.thepexcel.com/excel-running-total/>

| วิธี                         | จุดเด่น                                                | เหมาะกับกรณี                                 |
|------------------------------|--------------------------------------------------------|----------------------------------------------|
| 1.1 SUM ปกติ                 | เข้าใจง่าย มือใหม่ใช้ได้ทันที                          | ตารางเรียงลำดับแน่นอน                        |
| 1.2 SUM + [@Amount]          | ลากสูตรอัตโนมัติ, อ่านง่าย                             | ใช้กับ Table ได้ดี และประสิทธิภาพสูง         |
| 1.3 SUMIFS                   | ไม่พึ่งลำดับแถว, คงผลลัพธ์เดิมไว้ได้ แม้มีการสลับลำดับ | ข้อมูลถูก Sort ใหม่แล้วไม่พัง                |
| 2. Pivot Table               | ไม่ต้องเขียนสูตร                                       | วิเคราะห์ข้อมูลแบบ เปลี่ยนมุมมองได้ สะดวก    |
| 3. SCAN (Array)              | เขียนครั้งเดียว คำนวณทั้งชุด                           | งานไม่อยู่ใน Table / ชอบสูตรใหม่             |
| 4. Python                    | เขียนสั้น ใช้ .cumsum()                                | เขียน Script/Advanced Data Analysis          |
| 5.1 Power Query (Filter)     | ขั้นตอนชัดเจน                                          | เหมาะในขั้นตอนข้อมูล ที่ข้อมูลไม่เยอะ        |
| 5.2 Power Query (Accumulate) | เร็วกว่าแบบ Filter                                     | เหมาะในขั้นตอนข้อมูล ใช้กับข้อมูลจำนวนมากได้ |
| 6.1 DAX Column               | ค่าเก็บคงที่ ดูรายงานเร็ว                              | สร้าง Data Model                             |
| 6.2 DAX Measure              | Dynamic ตาม Context                                    | วิเคราะห์ใน PowerPivot / Visual              |

|    |                    |       |             |
|----|--------------------|-------|-------------|
| B1 |                    | $f_x$ | =LEFT(A1,5) |
|    | A                  | B     | C           |
| 1  | Learn Excel Easily | Learn |             |
| 2  |                    |       |             |

**=Left(text, num\_char)**

|    |                    |       |              |
|----|--------------------|-------|--------------|
| B1 |                    | $f_x$ | =RIGHT(A1,5) |
|    | A                  | B     | C            |
| 1  | Learn Excel Easily | asily |              |
| 2  |                    |       |              |

**=Right(text, num\_char)**

**=FIND(find\_text, within\_text,[start\_num])**

|    |                    |    |   |                  |  |
|----|--------------------|----|---|------------------|--|
| B1 |                    | fx |   | =FIND("Easi",A1) |  |
|    | A                  | B  | C | D                |  |
| 1  | Learn Excel Easily | 13 |   |                  |  |
| 2  |                    |    |   |                  |  |

Lab2.1

|    |                    |       |              |
|----|--------------------|-------|--------------|
| B1 |                    | $f_x$ | =MID(A1,7,5) |
|    | A                  | B     | C            |
| 1  | Learn Excel Easily | Excel |              |
| 2  |                    |       |              |

**= MID(text, start\_char, num\_chars)**

|    |                    |       |          |
|----|--------------------|-------|----------|
| B1 |                    | $f_x$ | =LEN(A1) |
|    | A                  | B     | C        |
| 1  | Learn Excel Easily | 18    |          |
| 2  |                    |       |          |

**= LEN(text)**

## Text to Number

=VALUE("10")

=VALUE("10.11")

## Number To Text

=TEXT(1234.567,"0")

=TEXT(1234.567,"\$#,##0.00")

=TEXT(0.285,"0.0%")

=TEXT(1234,"00000000")

=TEXT(TODAY(),"MM/DD/YY")

=TEXT(TODAY(),"DDDD")

=TEXT(NOW(),"H:MM AM/PM")

# Excel & Barcode

- Create Barcode 39 by Excel
- 128 Barcode
- Generate DataMatrix
- Power BI

Description CLASSICI  
TORTELLONIALLA RICOTTA E SPINACI

Article Code  
RAG040

Weight  
250g

GTIN number  
09012348410268

N° Pck  
2

Production date  
16/02/10

Lott no  
012345



(01)09012348410268(15)100816(10)012345

## GS1 Application Identifiers

| AI | Description                     | Format   |
|----|---------------------------------|----------|
| 01 | Global Trade Item Number (GTIN) | N2+N14   |
| 10 | Batch or lot number             | N2+X..20 |
| 15 | Best Before date (YYMMDD)       | N2+N6    |
| 17 | Expiration date (YYMMDD)        | N2+N6    |
| 21 | Serial number                   | N2+X..20 |

**UDI**



(01)00884838047693(21)XX01234567

**REF**

989605386721

SVC HW A.1



2014-03-13

**SN**



XX01234567



Made in USA of Domestic and Foreign Components  
Label PN 453561824762



123450036789



5 ตัว  
รหัสสินค้า  
ID



3 ตัว  
จำนวน  
Amount



4 ตัว  
รหัสจัดส่ง  
Shipment ID

(01) GTIN รหัสสินค้า

(17) วันหมดอายุ YYMMDD

(10) รหัสการผลิต

# Lab\_2\_2

| A1 |              |                                                                                                                                                                                                                                                    |                   |                   |
|----|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|
|    | A            | B                                                                                                                                                                                                                                                  | C                 | D                 |
| 1  |              | <div> <div>123450036789</div> <div> <div>5 ตัว</div> <div>รหัสสินค้า</div> <div>ID</div> </div> <div> <div>3 ตัว</div> <div>จำนวน</div> <div>Amount</div> </div> <div> <div>4 ตัว</div> <div>รหัสจัดส่ง</div> <div>Shipment ID</div> </div> </div> |                   |                   |
| 2  | ความยาว      | 5                                                                                                                                                                                                                                                  | 3                 | 4                 |
| 3  |              | ID                                                                                                                                                                                                                                                 | Amount            | ShipmentID        |
| 4  | 123450036789 | 12345                                                                                                                                                                                                                                              | 003               | 6789              |
| 5  |              | =LEFT(A4,5)                                                                                                                                                                                                                                        | =MID(A4,6,3)      | =RIGHT(A4,4)      |
| 6  |              |                                                                                                                                                                                                                                                    |                   |                   |
| 7  |              | =LEFT(A8,\$B\$2)                                                                                                                                                                                                                                   | =MID(A8,6,\$C\$2) | =RIGHT(A8,\$D\$2) |
| 8  | 123450036789 | 12345                                                                                                                                                                                                                                              | 003               | 6789              |
| 9  | 912345003678 | 91234                                                                                                                                                                                                                                              | 500               | 3678              |
| 10 | 891234500367 | 89123                                                                                                                                                                                                                                              | 450               | 0367              |
| 11 | 789123450036 | 78912                                                                                                                                                                                                                                              | 345               | 0036              |

# Lab2.3



(01)35019315001974(17)120300(10)1234567



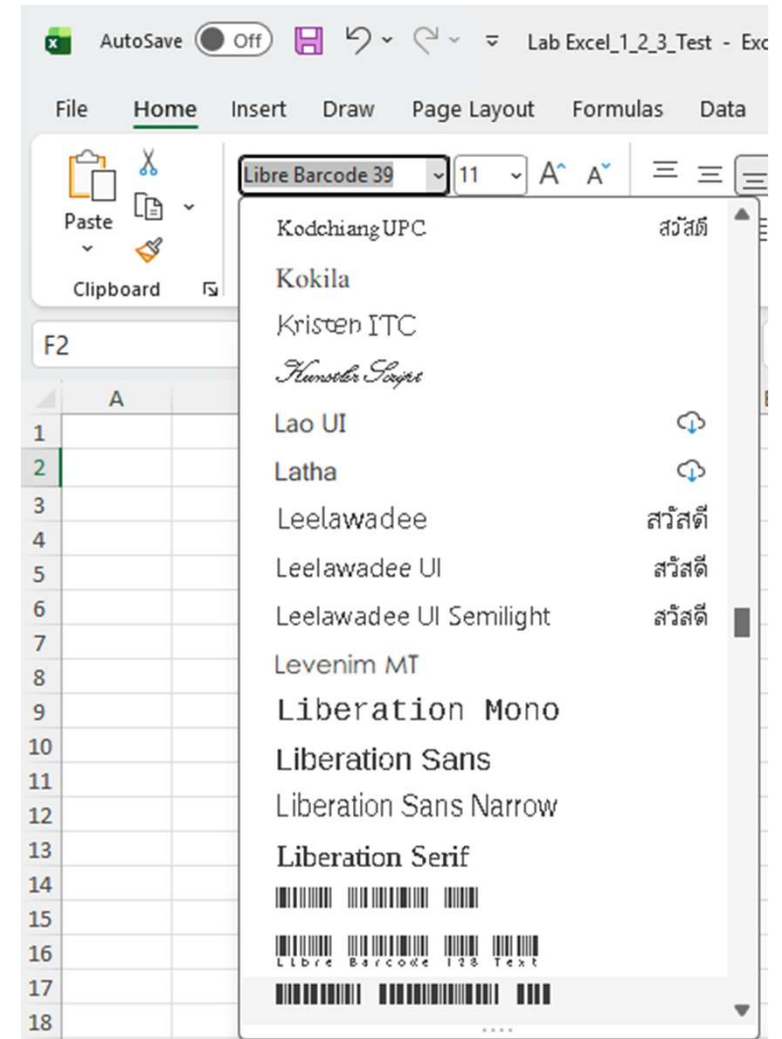
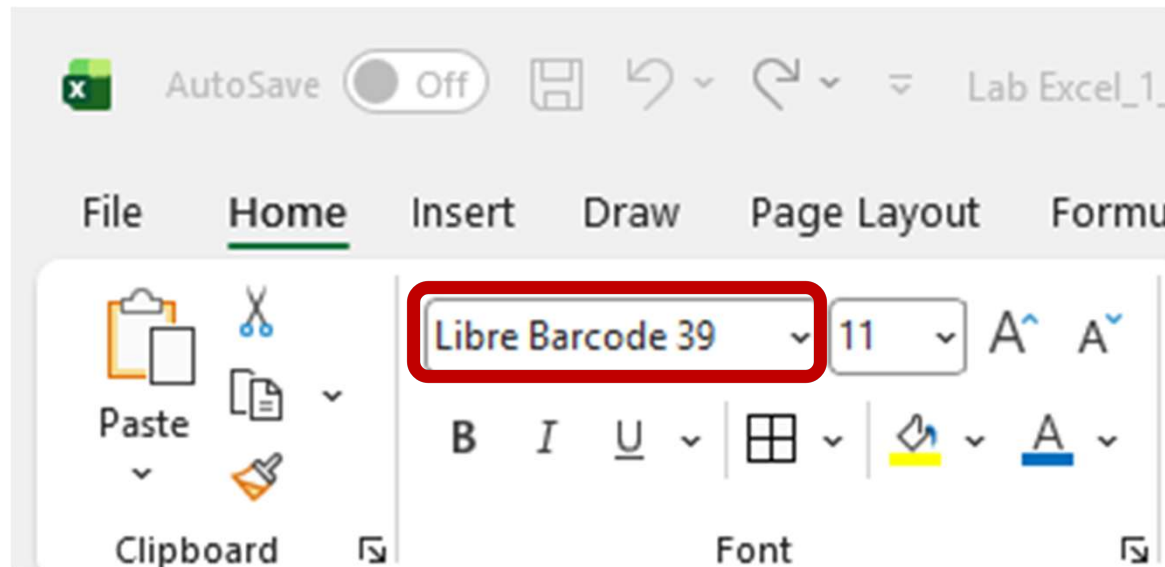
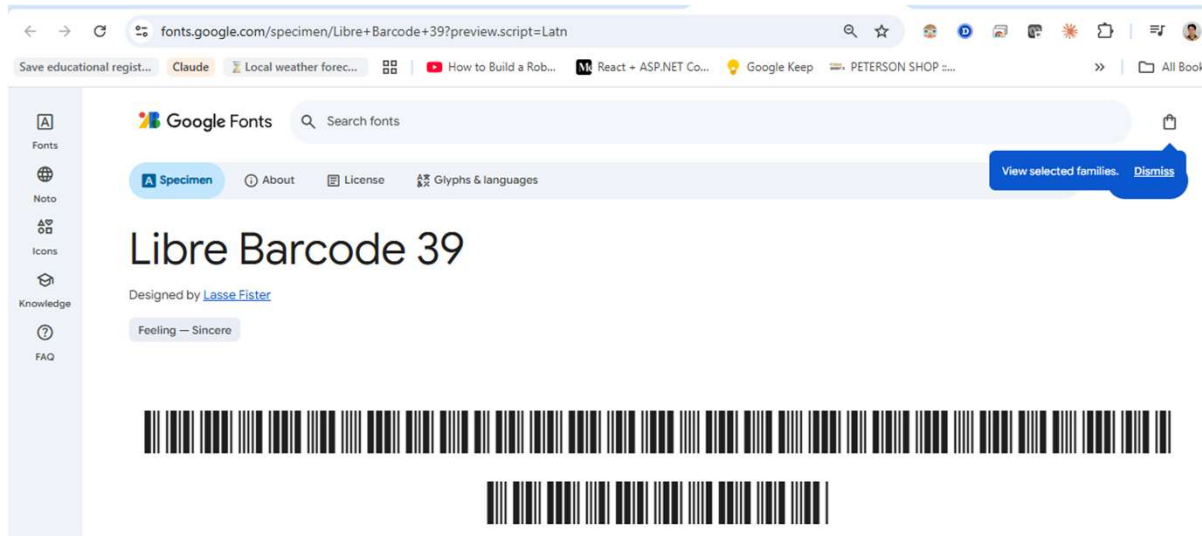
(01) GTIN รหัสสินค้า  
(17) วันหมดอายุ YYMMDD  
(10) รหัสการผลิต

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 1 | 3 | 5 | 0 | 1 | 9 | 3 | 1 | 5  | 0  | 0  | 1  | 9  | 7  | 4  | 1  | 7  | 1  | 2  | 0  | 3  | 0  | 0  | 1  | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

|    | A                                         | B | C              | D                   | E     | F      |
|----|-------------------------------------------|---|----------------|---------------------|-------|--------|
| 4  | ความยาว                                   |   | 33             | =LEN(A3)            | LEN   |        |
| 5  | ตัดซ้าย 2 ตัว                             |   | 01             | =LEFT(A3,2)         | LEFT  |        |
| 6  | รหัสสินค้า เริ่มตำแหน่งที่ 3 จำนวน 14 ตัว |   | 35019315001974 | =MID(A3,3,14)       | MID   |        |
| 7  | วันหมดอายุ 6 ตัวระหว่าง ตำแหน่ง 19 ถึง 24 |   | 200315         | =MID(A3,19,24-19+1) | MID   | YYMMDD |
| 8  | รหัสการผลิต ตัดข้อความจากขวา 7 ตัว        |   | 1234567        | =RIGHT(A3,7)        | RIGHT |        |
| 9  | ปี YY                                     |   | 2020           | =2000 + LEFT(C7,2)  | DD    |        |
| 10 | เดือน MM                                  |   | 03             | =MID(C7,3,2)        | MM    |        |
| 11 | วันที่ DD                                 |   | 15             | =RIGHT(C7,2)        | YY    |        |
| 12 | วันที่หมดอายุ                             |   | 15/3/2020      | =DATE(C9, C10, C11) | Date  |        |

## [Downlodin Libre Barcode 39 Font](https://fonts.google.com/specimen/Libre+Barcode+39?preview.script=Latn)

<https://fonts.google.com/specimen/Libre+Barcode+39?preview.script=Latn>



# Lab 3 รหัสบาร์โค้ด

## Lab3.1

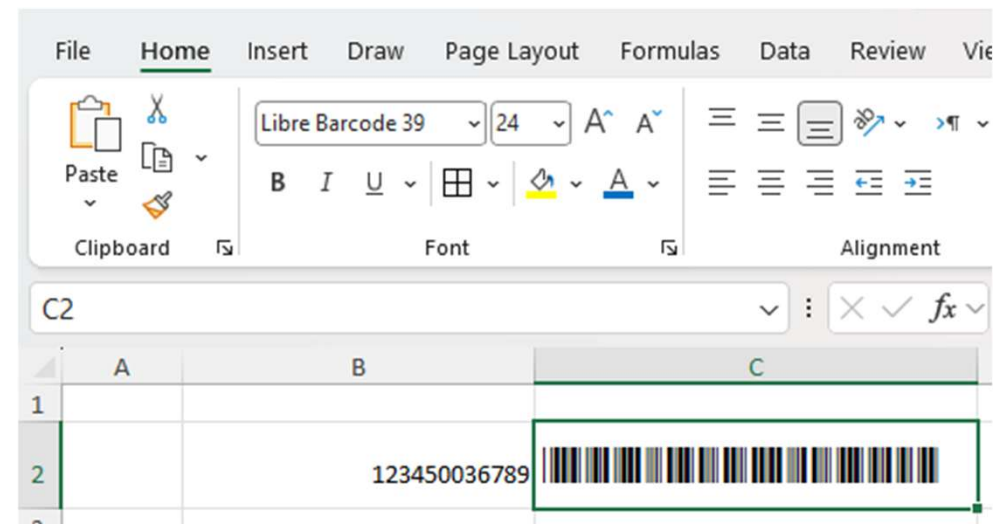
- การสร้างรหัส บาร์โค้ด
- การวิเคราะห์ข้อมูลในรหัสบาร์โค้ด

**\*123450036789\***

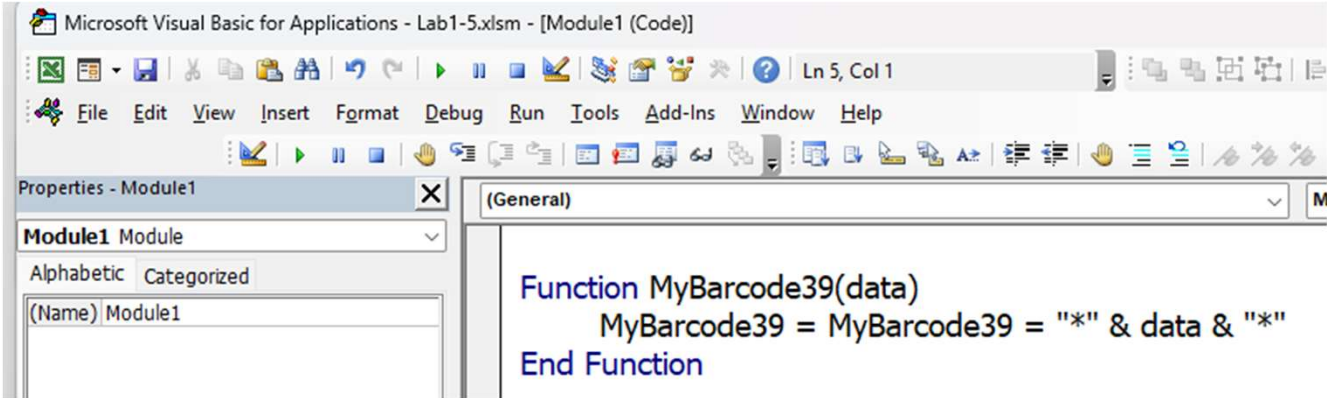
**= "\*" & B2 & "\*"**



|   | A | B            | C              |
|---|---|--------------|----------------|
| 1 |   |              |                |
| 2 |   | 123450036789 | *123450036789* |



# Lab3.1



|   | A | B            | C              |
|---|---|--------------|----------------|
| 1 |   |              |                |
| 2 |   | 123450036789 | *123450036789* |
| 3 |   |              | *123450036789* |

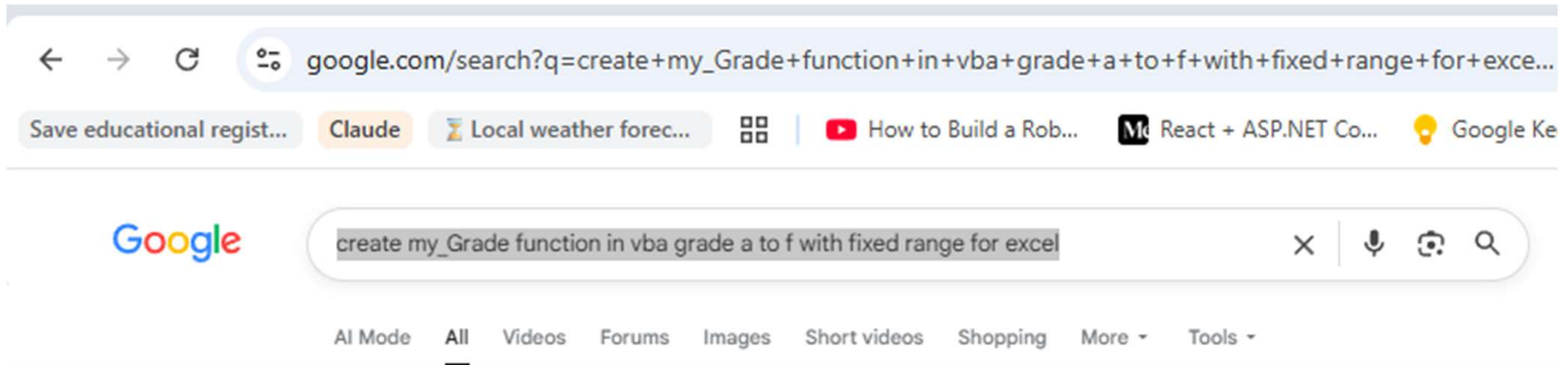


## Lab3.2

Ask AI in Google Search :

create my\_Grade function in vba grade a to f  
with fixed range for excel

<https://share.google/aimode/vlOy0tElkXqPQaVfe>



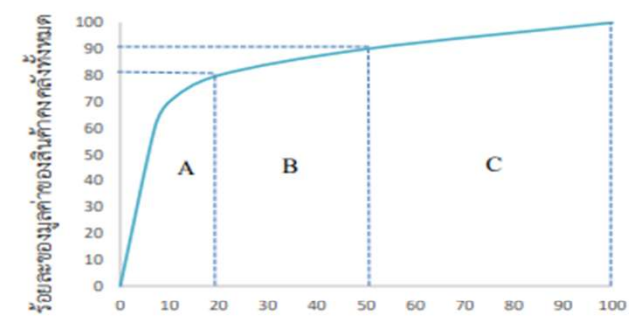
|   | A       | B          | C     | D     |
|---|---------|------------|-------|-------|
| 1 | Subject | NL2317     |       |       |
| 2 |         |            |       |       |
| 3 | ID      | NAME       | Score | Grade |
| 4 | 1       | นักศึกษา 1 | 75    |       |
| 5 | 2       | นักศึกษา 2 | 82    |       |
| 6 | 3       | นักศึกษา 3 | 50    |       |
| 7 | 4       | นักศึกษา 4 | 45    |       |
| 8 | 5       | นักศึกษา 5 | 54    |       |
| 9 | 6       | นักศึกษา 6 | 45    |       |

```
Function my_Grade(score As Double) As String
    Select Case score
        Case Is < 0
            my_Grade = "Invalid"
        Case Is < 60
            my_Grade = "F"
        Case Is < 70
            my_Grade = "D"
        Case Is < 80
            my_Grade = "C"
        Case Is < 90
            my_Grade = "B"
        Case Is <= 100
            my_Grade = "A"
        Case Else
            my_Grade = "Invalid"
    End Select
End Function
```

# Lab 3 3

ตารางที่ 1 การจำแนกกลุ่มสินค้าคงคลังตามการวิเคราะห์แบบเอบีซี (ABC Analysis)

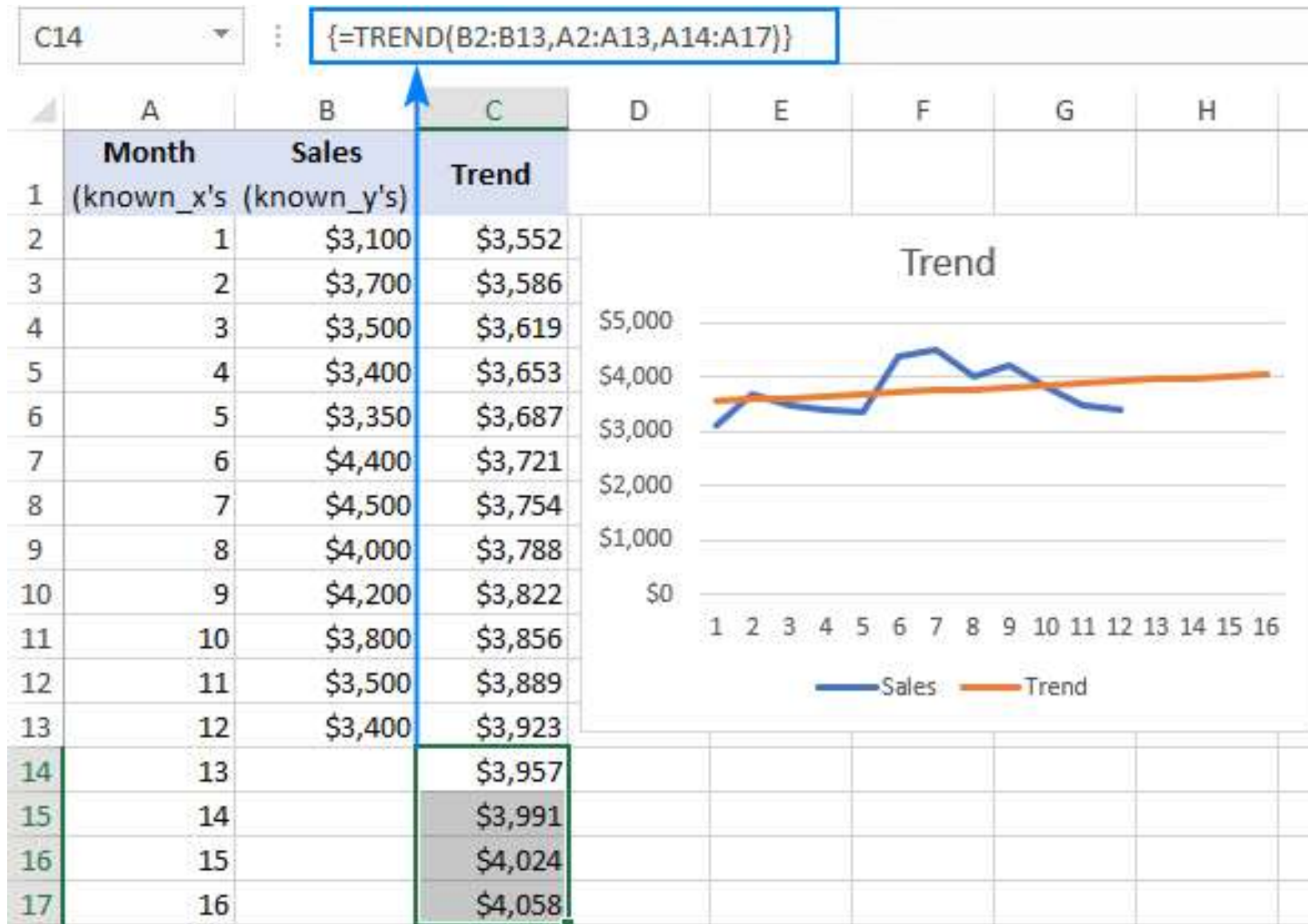
| กลุ่ม | ร้อยละของมูลค่าสินค้าคงคลังทั้งหมด | ร้อยละของปริมาณการใช้สินค้าคงคลังทั้งหมด |
|-------|------------------------------------|------------------------------------------|
| A     | 70-80                              | 10-20                                    |
| B     | 15-20                              | 30-40                                    |
| C     | 5-10                               | 40-50                                    |



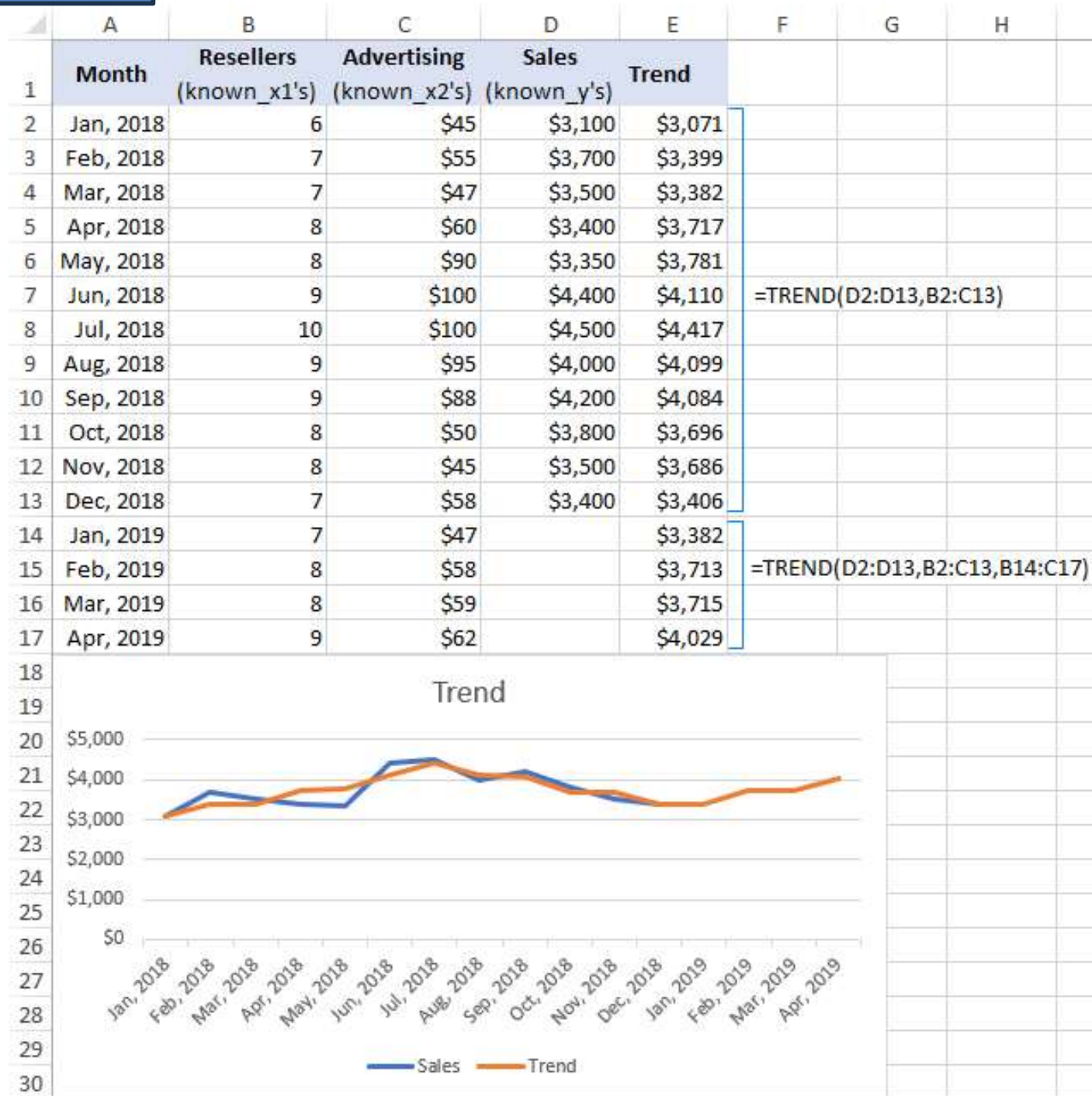
ตารางที่ 4 ร้อยละของปริมาณการใช้สินค้าคงคลังทั้งหมดและร้อยละของมูลค่าสินค้าคงคลังแต่ละรายการ

| ลำดับที่ | รายการสินค้าคงคลัง(รหัสสินค้า) | ปริมาณการใช้สินค้าคงคลัง (หน่วย) |         |        |            | มูลค่าสินค้าคงคลัง (บาท) |           |        |            | กลุ่ม |
|----------|--------------------------------|----------------------------------|---------|--------|------------|--------------------------|-----------|--------|------------|-------|
|          |                                | ต่อปี                            | สะสม    | ร้อยละ | ร้อยละสะสม | ต่อปี                    | สะสม      | ร้อยละ | ร้อยละสะสม |       |
| 1        | N8                             | 17,300                           | 17,300  | 13.15  | 13.15      | 432,500                  | 432,500   | 38.15  | 38.15      | A     |
| 2        | G7                             | 14,000                           | 31,300  | 10.64  | 23.79      | 210,000                  | 642,500   | 18.52  | 56.67      | A     |
| 3        | I9                             | 15,700                           | 47,000  | 11.93  | 35.72      | 188,400                  | 830,900   | 16.62  | 73.29      | A     |
| 4        | P3                             | 15,500                           | 62,500  | 11.78  | 47.50      | 108,500                  | 939,400   | 9.57   | 82.86      | B     |
| 5        | C6                             | 2,360                            | 64,860  | 1.79   | 49.29      | 70,800                   | 1,010,200 | 6.24   | 89.11      | B     |
| 6        | B8                             | 6,260                            | 71,120  | 4.76   | 54.05      | 32,865                   | 1,043,065 | 2.90   | 92.00      | C     |
| 7        | L9                             | 7,200                            | 78,320  | 5.47   | 59.52      | 32,400                   | 1,075,465 | 2.86   | 94.86      | C     |
| 8        | M2                             | 17,200                           | 95,520  | 13.07  | 72.59      | 25,800                   | 1,101,265 | 2.28   | 97.14      | C     |
| 9        | A1                             | 500                              | 96,020  | 0.38   | 72.97      | 10,000                   | 1,111,265 | 0.88   | 98.02      | C     |
| 10       | D3                             | 2,300                            | 98,320  | 1.75   | 74.72      | 6,900                    | 1,118,165 | 0.61   | 98.63      | C     |
| 11       | O9                             | 15,700                           | 114,020 | 11.93  | 86.65      | 5,495                    | 1,123,660 | 0.48   | 99.11      | C     |
| 12       | V4                             | 5,060                            | 119,080 | 3.85   | 90.50      | 4,301                    | 1,127,961 | 0.38   | 99.49      | C     |
| 13       | X2                             | 1,500                            | 120,580 | 1.14   | 91.64      | 3,000                    | 1,130,961 | 0.26   | 99.76      | C     |
| 14       | S5                             | 11,000                           | 131,580 | 8.36   | 100.00     | 2,750                    | 1,133,711 | 0.24   | 100.00     | C     |

# Lab 4.1



# Lab 4.2





# Lab 4.3

<https://www.ablebits.com/office-addins-blog/moving-average-excel/>

|    |       |               |                |                 |   |
|----|-------|---------------|----------------|-----------------|---|
| C4 |       | =SUM(B2:B4)/3 |                | =AVERAGE(B2:B4) |   |
|    | A     | B             | C              | D               | E |
| 1  | Month | Temp. °F      | Moving Average |                 |   |
| 2  | Jan   | 39            |                |                 |   |
| 3  | Feb   | 42            |                |                 |   |
| 4  | Mar   | 50            | 44             | 44              |   |
| 5  | Apr   | 60            | 51             | 51              |   |
| 6  | May   | 71            | 60             | 60              |   |
| 7  | Jun   | 79            | 70             | 70              |   |
| 8  | Jul   | 85            | 78             | 78              |   |
| 9  | Aug   | 81            | 82             | 82              |   |
| 10 | Sep   | 76            | 81             | 81              |   |

# Lab 4.4

`=AVERAGE(OFFSET(first cell, COUNT(entire range)-N,0,N,1))`

|    |       |       |                                                          |   |   |   |          |
|----|-------|-------|----------------------------------------------------------|---|---|---|----------|
| G1 |       | :     | <code>=AVERAGE(OFFSET(B2,COUNT(B2:B100)-3,0,3,1))</code> |   |   |   |          |
|    | A     | B     | C                                                        | D | E | F | G        |
| 1  | Month | Sales | Average for last 3 months:                               |   |   |   | \$276.67 |
| 2  | Jan   | \$200 |                                                          |   |   |   |          |
| 3  | Feb   | \$320 |                                                          |   |   |   |          |
| 4  | Mar   | \$180 |                                                          |   |   |   |          |
| 5  | Apr   | \$450 |                                                          |   |   |   |          |
| 6  | May   | \$360 |                                                          |   |   |   |          |
| 7  | Jun   | \$400 |                                                          |   |   |   |          |
| 8  | Jul   | \$240 |                                                          |   |   |   |          |
| 9  | Aug   | \$280 | (240+280+310)/3=276.67                                   |   |   |   |          |
| 10 | Sep   | \$310 |                                                          |   |   |   |          |
| 11 | Oct   |       |                                                          |   |   |   |          |
| 12 | Nov   |       |                                                          |   |   |   |          |
| 13 | Dec   |       |                                                          |   |   |   |          |

$(240+280+310)/3=276.67$

|   | A | B                                | C    | D    | E     | F |
|---|---|----------------------------------|------|------|-------|---|
| 1 |   |                                  |      |      |       |   |
| 2 |   |                                  |      |      |       |   |
| 3 |   | นก                               | หมู  | หมา  | หมี   |   |
| 4 |   | แมว                              | ช้าง | ม้า  | ลา    |   |
| 5 |   | ปลา                              | สิง  | เสือ | สิงโต |   |
| 6 |   |                                  |      |      |       |   |
| 7 |   |                                  |      |      |       |   |
| 8 |   | สิงโต                            |      |      |       |   |
| 9 |   | <code>=OFFSET(C3,2,3,1,1)</code> |      |      |       |   |

<https://www.ablebits.com/office-addins-blog/moving-average-excel/>



## Lab 4.5

```
=AVERAGE(OFFSET(first cell,0,COUNT(range)-N,1,N,))
```

The screenshot shows an Excel spreadsheet with the following data:

|   | A                         | B     | C     | D     | E     | F        | G     | H     | I     | J     | K   | L   |
|---|---------------------------|-------|-------|-------|-------|----------|-------|-------|-------|-------|-----|-----|
| 1 | Month                     | Jan   | Feb   | Mar   | Apr   | May      | Jun   | Jul   | Aug   | Sep   | Oct | Nov |
| 2 | Sales                     | \$200 | \$320 | \$180 | \$450 | \$360    | \$400 | \$240 | \$280 | \$310 |     |     |
| 3 |                           |       |       |       |       |          |       |       |       |       |     |     |
| 4 | Average for last 3 months |       |       |       |       | \$276.67 |       |       |       |       |     |     |

The formula bar shows the formula: `=AVERAGE(OFFSET(B2, 0, COUNT(B2:N2)-3,1,3))`. An orange arrow points from the formula bar to cell E4, indicating that the formula is applied to that cell.